PERIMETERS OF PRIMITIVE HERONIAN ISOSCELES TRIANGLES: A FORMAL PROOF THAT $\mathrm{A308222}(n) = 2 \cdot \mathrm{A096033}(n+2)$

MATH0

Abstract. Let $a(n)$ be the n-th smallest perimeter of a primitive Heronian isosceles triangle (OEIS A308222), and let $\mathrm{A096033}(n)$ be the n-th smallest difference between a leg and the hypotenuse of a primitive Pythagorean triangle. We prove that $a(n) = 2 \cdot \mathrm{A096033}(n+2)$ for every $n \geq 1$, an identity that P. Kagey recorded as a conjecture in 2019 and that the entry still marks as conjectured. The identity is a short consequence of the classical description of primitive isosceles Heronian triangles as two copies of a primitive Pythagorean triangle glued along a leg (Carlson 1970, corrected by Singmaster 1973), and a Mathematics Stack Exchange answer linked from the entry derives the relevant semiperimeters; we therefore classify our result as the first formal proof of a known result. Our proof shows directly that $2m$ is a perimeter if and only if $m \geq 3$ is an odd square or twice a square, and transfers this to the two sequences by a counting argument. It is verified in Lean 4 against a statement that math0 formalized from the OEIS entry, reviewed for fidelity, and frozen before the proof attempt.

1. Introduction

A triangle is *Heronian* if its side lengths and its area are integers, and *primitive* if its three side lengths have no common divisor greater than 1. The OEIS entry A308222 [8], contributed by Peter Kagey on May 16, 2019, is "Numbers that are the perimeter of a primitive Heronian isosceles triangle": $16, 18, 36, 50, 64, 98, 100, 144, \ldots$, starting with the triangles $(5, 5, 6)$ and $(5, 5, 8)$. Its FORMULA section relates it to A096033 [7], "Difference between leg and hypotenuse in primitive Pythagorean triangles": $1, 2, 8, 9, 18, 25, 32, 49, 50, \ldots$ (Beedassy, 2004). The conjecture is that the perimeters are twice the leg–hypotenuse differences, with an index shift of two; Theorem 2 proves it.

Prior work. The structure behind the identity is classical. Carlson [1] proved that every Heronian triangle is, up to scaling, two Pythagorean triangles joined along a common side. His Lemma 2 states that a primitive Heronian triangle is isosceles if and only if its sides are given by his general formulas with the two Pythagorean triangles taken equal, that is, $(u^2 + v^2,\ u^2 + v^2,\ 2(u^2 - v^2))$ or $(u^2 + v^2,\ u^2 + v^2,\ 4uv)$, or a reduction of one of these; in other words, the triangle is two copies of one Pythagorean triangle. Singmaster [9] pointed out errors in Carlson's paper (notably the use of the form $u^2 + v^2, u^2 - v^2, 2uv$ for all Pythagorean triangles), remarked that Lemma 2 is unclearly stated, and gave a corrected form of it: the isosceles Heronian triangles are exactly $a(u^2 + v^2), a(u^2 + v^2), 4auv$ and $a(u^2 + v^2), a(u^2 + v^2), 2a(u^2 - v^2)$, and such a triangle is primitive if and only if $a = 1$, $\gcd(u, v) = 1$ and $u \not\equiv v \pmod 2$. The same parametrization appears in the Wikipedia article [10], which the entry links. On May 17, 2019, the day after creating the entry, Kagey asked on Mathematics Stack Exchange [4] for a characterization of the semiperimeters of (not necessarily primitive) isosceles Heronian triangles. An answer by the user "George" [3], linked from A308222, quotes Carlson's lemma and concludes that the semiperimeters of the primitive ones are $2u^2$ and $(u + v)^2$. The answer states only this necessity direction; it does not state that every such value occurs, nor the identity with

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A096033. That A096033 consists of the odd squares and the numbers $2j^2$ is a comment in that entry (A. Weimholt, 2010).

From Singmaster's form of Lemma 2 the conjecture follows in a few lines (Remark 12). Accordingly we classify our result as the *first formal proof of a known result*. The math0 referees found no explicit statement and proof of the identity itself in the literature or in the cross-referenced OEIS entries, and no prior formalization; the current revision of A308222 (#20, August 12, 2022) still marks it "(conjectured)". Our proof (Section 3) works with the perimeter directly: it reduces the area condition to "$m(2s-m)$ is a square" and handles the two parities of m by coprimality arguments, with explicit legs and Bézout certificates for the converse.

2. Statement

Definition 1. For $x, y, z \in \mathbb{N}$ let $H(x,y,z) = (x+y+z)(-x+y+z)(x-y+z)(x+y-z) \in \mathbb{Z}$. By Heron's formula, $H(x,y,z) = 16 \cdot \text{area}^2$ for a triangle with sides x, y, z.

(1) $\mathcal{P}$ is the set of $P \in \mathbb{N}$ for which there are $x, y, z \in \mathbb{N}$ with $x+y+z = P$ such that: $x < y+z$, $y < x+z$, $z < x+y$; $16A^2 = H(x,y,z)$ for some $A \in \mathbb{N}$; $\gcd(x,y,z) = 1$; and two of x, y, z are equal.
(2) $\mathcal{L}$ is the set of $d \in \mathbb{N}$ for which there are $x, y, z \in \mathbb{N}$ with $x, y \geq 1$, $x^2 + y^2 = z^2$, $\gcd(x,y) = 1$ and $z = x + d$.
(3) $a(n)$ and A096033(n), for $n \geq 1$, are the n-th smallest elements of $\mathcal{P}$ and of $\mathcal{L}$.

Both sets are infinite (Lemma 6 and Theorem 3 below), so $a(n)$ and A096033(n) are defined for every $n \geq 1$.

Theorem 2 (Kagey's conjecture)**.** *For every $n \geq 1$, $a(n) = 2 \cdot \text{A096033}(n+2)$.*

For example $a(1) = 16 = 2 \cdot 8$ and $a(2) = 18 = 2 \cdot 9$. For $n = 0$ the shifted formula would read $a(0) = 2 \cdot \text{A096033}(2) = 4$, but $4 \notin \mathcal{P}$; the conjecture is stated for $n \geq 1$, the range of the entry. The proof also gives the set of perimeters explicitly:

Theorem 3. $\mathcal{P} = \{2d : d \in \mathcal{L},\ d \geq 3\} = \{2k^2 : k \geq 3 \text{ odd}\} \cup \{4j^2 : j \geq 2\}$.

The conjecture in the OEIS. The FORMULA line of A308222 [8] was added by Kagey on May 17, 2019 (revision #9 of the entry), a few hours after the Stack Exchange answer [3] appeared and before the entry was first approved on May 23, 2019. It reads verbatim:

```
a(n) = 2*A096033(n+2) (conjectured).
```

The entry's comment line reads: "A primitive Heronian triangle is a triangle with integer sides and area, where the side lengths do not share a common divisor." Both entries have offset 1.

The Lean statement. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A308222.lean`, reviewed for fidelity against the OEIS entries, and frozen (SHA-256 `99173e00...b7c8`) before any proof attempt. Its definitions (without their docstrings) and the statement (with its docstring) are, verbatim, in the namespace `Math0.Statements.A308222`:

```
def heron (x y z : ℤ) : ℤ := (x + y + z) * (-x + y + z) * (x - y + z) * (x + y - z)
def IsHeronian (x y z : ℕ) : Prop :=
  x < y + z ∧ y < x + z ∧ z < x + y ∧ ∃ A : ℕ, 16 * (A : ℤ) ^ 2 = heron x y z
def IsPrimitive (x y z : ℕ) : Prop := Nat.gcd (Nat.gcd x y) z = 1
def IsIsosceles (x y z : ℕ) : Prop := x = y ∨ y = z ∨ x = z
def IsPerimeter (P : ℕ) : Prop :=
  ∃ x y z : ℕ, IsHeronian x y z ∧ IsPrimitive x y z ∧ IsIsosceles x y z ∧ x + y + z = P
noncomputable def a (n : ℕ) : ℕ := Nat.nth IsPerimeter (n - 1)

def IsLegHypDiff (d : ℕ) : Prop :=
  ∃ x y z : ℕ, 0 < x ∧ 0 < y ∧ x ^ 2 + y ^ 2 = z ^ 2 ∧ Nat.gcd x y = 1 ∧ z = x + d
```

```
noncomputable def A096033 (n : ℕ) : ℕ := Nat.nth IsLegHypDiff (n - 1)

/-- Conjecture (Kagey, A308222): for every `n ≥ 1`, the `n`-th smallest perimeter of a primitive
Heronian isosceles triangle is twice the `(n + 2)`-th smallest difference between a leg and the
hypotenuse of a primitive Pythagorean triangle: `a(n) = 2 · A096033(n + 2)`. -/
def statement : Prop := ∀ n : ℕ, 1 ≤ n → a n = 2 * A096033 (n + 2)
```

How the Lean definitions encode the entries. `IsPerimeter` is the set $\mathcal{P}$ and `IsLegHypDiff` is the set $\mathcal{L}$ of Definition 1. Both are defined from the names of the entries, not from any formula. Heron's product is computed in $\mathbb{Z}$, and the strict triangle inequalities exclude degenerate triangles. Mathlib's `Nat.nth p k` is the k-th smallest element of $\{x : p\,x\}$ (counting from 0) when this set is infinite, so `a n` and `A096033 n` are the n-th smallest elements for $n \geq 1$, matching offset 1. Each perimeter is listed once, as in the entry: 50 comes from both $(13, 13, 24)$ and $(16, 17, 17)$. The statement is the FORMULA line for all $n \geq 1$, with no further hypotheses.

The statement file checks the definitions against the data. Since `Nat.nth` is not computable, it makes both predicates decidable through bounded searches that are proved equivalent to them: `perimB_iff` rests on the reduction of Lemma 5 below, with a certified integer square root, and `legHypB_iff` on the characterization of $\mathcal{L}$ in Lemma 6. With these, `a_data` proves by `decide +kernel` that $a(1), \ldots, a(46)$ are the 46 listed terms $16, \ldots, 3136$. The search below 3137 is split into seven kernel checks (`chunk_1`, …, `chunk_7`) to bound memory. `A096033_data` proves the 52 listed terms of A096033, $1, \ldots, 1849$, in the same way. These checks are part of the frozen statement file and were done before the proof attempt.

3. Proof

For a predicate p on $\mathbb{N}$ write $\mathrm{count}_p(N) = \#\{k < N : p(k)\}$ and $\mathrm{nth}_p(k)$ for the k-th smallest element of $\{x : p(x)\}$, counting from 0 (Mathlib's `Nat.count` and `Nat.nth`). Then $a(n) = \mathrm{nth}_{\mathcal{P}}(n-1)$ and $\mathrm{A096033}(n) = \mathrm{nth}_{\mathcal{L}}(n-1)$. The first two lemmas are proved in the statement file, where they serve to decide the predicates; the rest of the section is the proof file.

Definition 4. For $m, s \in \mathbb{N}$, $\mathrm{LegOK}(m, s)$ means: $m < 2s$, $s < m$, s is odd, $\gcd(s, m) = 1$, and $m(2s - m)$ is a perfect square.

Lemma 5 (`isPerimeter_iff_legOK`). *$P \in \mathcal{P}$ if and only if $P = 2m$ is even and $\mathrm{LegOK}(m, s)$ holds for some s. In that case $(s, s, 2(m-s))$ is a primitive Heronian triangle with perimeter P.*

Proof. Since H is symmetric in its arguments, every primitive Heronian isosceles triangle can be written with legs s, s and base b (`isPerimeter_iff`). The triangle inequalities become $0 < b < 2s$, primitivity becomes $\gcd(s, b) = 1$, and $H(s, s, b) = (2s + b)\, b^2\, (2s - b)$ (`heron_iso`). If b were odd, all three factors would be odd, while $H = 16A^2$ is even; so $b = 2x$ is even (`IsoOK.base_even`), and then s is odd because $\gcd(s, b) = 1$ (`IsoOK.leg_odd`). Put $m = s + x$, so that $P = 2s + 2x = 2m$ and $s - x = 2s - m$. Then

$$H(s, s, 2x) = (2s + 2x)(2x)^2(2s - 2x) = 16\, x^2\, m\, (2s - m),$$

so the area condition is $A^2 = x^2 \cdot m(2s - m)$. As $x \geq 1$, this equation has a solution $A \in \mathbb{N}$ if and only if $m(2s - m)$ is a square: if $A^2 = x^2 K$ then $x^2 \mid A^2$, so $x \mid A$ and $K = (A/x)^2$; conversely $A = xh$ works if $K = h^2$ (`exists_sq_mul_iff`). Moreover $0 < b$ iff $s < m$, $b < 2s$ iff $m < 2s$, and, for odd s, $\gcd(s, 2x) = 1$ iff $\gcd(s, x) = 1$ iff $\gcd(s, s + x) = \gcd(s, m) = 1$ (`isOK_iff_legOK`). Reading these equivalences backwards gives the converse. □

Lemma 6 (`isLegHypDiff_iff`). *$d \in \mathcal{L}$ if and only if d is an odd square or $d = 2j^2$ with $j \geq 1$.*

Proof. Let $x^2 + y^2 = z^2$ with $x, y \geq 1$, $\gcd(x, y) = 1$ and $z = x + d$. By the classification of primitive Pythagorean triples (Mathlib's `PythagoreanTriple.coprime_classification'`), exactly one leg is even, and there are coprime integers p, q of opposite parity with $z = p^2 + q^2$ and either $x = 2pq$, $y = p^2 - q^2$, or $x = p^2 - q^2$, $y = 2pq$. In the first case $d = (p - q)^2$ with $p - q$ odd. In the second case $d = 2q^2$, and $q \neq 0$ because $y > 0$. Conversely, for odd k the triple $(x, y, z) = (2k+2,\ k^2+2k,\ k^2+2k+2)$ satisfies $x^2+y^2 = z^2$ and $z-x = k^2$; with $k = 2t+1$, the identity $(t+1)x - y = 1$ shows $\gcd(x, y) = 1$. For $j \geq 1$ the triple $(2j+1,\ 2j^2+2j,\ 2j^2+2j+1)$ satisfies $x^2 + y^2 = z^2$ and $z - x = 2j^2$, and $(2j + 1)x - 2y = 1$. □

Consequently $\mathcal{L} \cap [0, 3) = \{1, 2\}$, and $\mathcal{L}$ is infinite because it contains $2(N + 1)^2$ for every N (`infinite_L`, in the proof file).

Lemma 7 (`sq_of_coprime_mul`)**.** *If $a, b \in \mathbb{N}$ are coprime and $ab = c^2$, then a is a perfect square.*

Proof. This is Mathlib's `exists_eq_pow_of_mul_eq_pow`, applied in $\mathbb{N}$, where the only unit is 1. □

Lemma 8 (`isLegHypDiff_of_legOK`)**.** *If* $\mathrm{LegOK}(m, s)$ *holds for some s, then $m \in \mathcal{L}$.*

Proof. Write $2s = m + e$ with $e \geq 1$, and let $h^2 = me$.

m odd. Then $\gcd(m, 2) = 1$ and $\gcd(m, s) = 1$, so $\gcd(m, 2s) = 1$, and $\gcd(m, e) = \gcd(m, m+e) = \gcd(m, 2s) = 1$. By Lemma 7, m is a square, and it is odd.

$m = 2m'$ even. Since $2s = 2m' + e$, $e = 2f$ is even and $s = m' + f$. As $m' \mid m$ and $\gcd(s, m) = 1$, $\gcd(m', s) = 1$, hence $\gcd(m', f) = \gcd(m', m' + f) = 1$. Now $h^2 = 4m'f$ is even, so $h = 2r$ and $r^2 = m'f$. By Lemma 7, $m' = k^2$, so $m = 2k^2$; and $k \geq 1$ because $m > s \geq 1$.

In both cases $m \in \mathcal{L}$ by Lemma 6. □

Lemma 9 (`legOK_of_isLegHypDiff`)**.** *If $m \in \mathcal{L}$ and $m \geq 3$, then* $\mathrm{LegOK}(m, s)$ *holds for some s.*

Proof. By Lemma 6 there are two cases.

m an odd square. Since $m \geq 3$, $m = (2u + 3)^2$ with $u \geq 0$. Take $s = 4(u + 1)^2 + 1$, which is odd. Then

$$2s - m = 8u^2 + 16u + 10 - (4u^2 + 12u + 9) = (2u + 1)^2 > 0, \qquad m - s = 4u + 4 > 0,$$

so $m < 2s$, $s < m$, and $m(2s - m) = \big((2u + 3)(2u + 1)\big)^2$. Finally $(u + 2)s - (u + 1)m = 1$, as both $(u + 2)(4u^2 + 8u + 5)$ and $(u + 1)(4u^2 + 12u + 9) + 1$ equal $4u^3 + 16u^2 + 21u + 10$; hence $\gcd(s, m) = 1$.

$m = 2j^2$, $j \geq 1$. Since $m \geq 3$, $j = v + 2$ with $v \geq 0$. Take $s = (v + 2)^2 + (v + 1)^2$, which is odd as a sum of squares of consecutive integers. Then $2s - m = 2(v + 1)^2 > 0$ and $m - s = (v+2)^2 - (v+1)^2 = 2v+3 > 0$, so $m < 2s$, $s < m$, and $m(2s-m) = \big(2(v+1)(v+2)\big)^2$. Finally $(2v+5)s - (2v+3)m = 1$, as both $(2v+5)(2v^2+6v+5)$ and $(2v+3)(2v^2+8v+8)+1$ equal $4v^3 + 22v^2 + 40v + 25$; hence $\gcd(s, m) = 1$. □

By Lemma 5, the legs of Lemma 9 give the triangles $(p^2 + 1,\ p^2 + 1,\ 4p)$ with $p = 2u + 2$ and $(q^2 + (q - 1)^2,\ q^2 + (q - 1)^2,\ 2(2q - 1))$ with $q = v + 2$; for $u = v = 0$ these are $(5, 5, 8)$ and $(5, 5, 6)$.

Corollary 10 (`isPerimeter_two_mul`, `not_isPerimeter_odd`)**.** *No odd number lies in $\mathcal{P}$, and for $n \geq 3$, $2n \in \mathcal{P}$ if and only if $n \in \mathcal{L}$.*

Proof. The first claim is part of Lemma 5. For $n \geq 3$, Lemma 5 says $2n \in \mathcal{P}$ iff $\mathrm{LegOK}(n, s)$ for some s, and Lemmas 8 and 9 show that this holds iff $n \in \mathcal{L}$. □

Proof of Theorem 3. By Lemma 5, $\mathcal{P}$ consists of even numbers $2m$ with some s satisfying $m/2 < s < m$ and s odd, which is impossible for $m \le 2$. Corollary 10 handles $m \ge 3$, and Lemma 6 describes $\mathcal{L}$. □

The proof of Theorem 2 in Lean does not use Theorem 3 as a set equality; it uses Corollary 10 and a counting identity.

Lemma 11 (`count_rel`)**.** *For every $n \ge 3$, $\mathrm{count}_{\mathcal{P}}(2n) + 2 = \mathrm{count}_{\mathcal{L}}(n)$.*

Proof. Induction on $n \ge 3$. For $n = 3$: $\mathcal{P} \cap [0, 6) = \emptyset$ and $\mathcal{L} \cap [0, 3) = \{1, 2\}$, so $0 + 2 = 2$. In Lean this case is checked by `decide`, which evaluates both counts with the certified decision procedures of the statement file. For the step from n to $n + 1$,

$$\mathrm{count}_{\mathcal{P}}(2n+2) = \mathrm{count}_{\mathcal{P}}(2n) + [2n \in \mathcal{P}] + [2n+1 \in \mathcal{P}] = \mathrm{count}_{\mathcal{P}}(2n) + [n \in \mathcal{L}]$$

by Corollary 10, while $\mathrm{count}_{\mathcal{L}}(n+1) = \mathrm{count}_{\mathcal{L}}(n) + [n \in \mathcal{L}]$. □

Proof of Theorem 2. Let $n \ge 1$ and $k = n - 1$, so that $a(n) = \mathrm{nth}_{\mathcal{P}}(k)$ and $\mathrm{A096033}(n+2) = \mathrm{nth}_{\mathcal{L}}(k+2) =: d$. Since $\mathcal{L}$ is infinite (Lemma 6), $d \in \mathcal{L}$ and $\mathrm{count}_{\mathcal{L}}(d) = k+2$ (`Nat.nth_mem_of_infinite`, `Nat.count_nth_of_infinite`). If $d \le 2$, then $\mathrm{count}_{\mathcal{L}}(d) \le \mathrm{count}_{\mathcal{L}}(2) = 1 < k + 2$ (the value $\mathrm{count}_{\mathcal{L}}(2) = 1$ is checked by `decide`); hence $d \ge 3$. By Corollary 10, $2d \in \mathcal{P}$, and by Lemma 11, $\mathrm{count}_{\mathcal{P}}(2d) = k$. An element x of a set with exactly k elements of the set below x is its k-th smallest element (`Nat.nth_count`), so $\mathrm{nth}_{\mathcal{P}}(k) = 2d$, that is, $a(n) = 2 \cdot \mathrm{A096033}(n+2)$. □

Remark 12 (derivation from the literature; not formalized)**.** By Singmaster's form of Carlson's Lemma 2 [9], the primitive isosceles Heronian triangles are $(u^2 + v^2, u^2 + v^2, 4uv)$ and $(u^2 + v^2, u^2 + v^2, 2(u^2 - v^2))$ with $u > v \ge 1$, $\gcd(u, v) = 1$ and $u \not\equiv v \pmod 2$. Their semiperimeters are $(u + v)^2$ and $2u^2$, as in [3]. Here $u + v$ is odd and at least 3, and every odd $k \ge 3$ occurs, with $(u, v) = ((k + 1)/2, (k - 1)/2)$; and $u \ge 2$, where every $u \ge 2$ occurs, with $v = u - 1$. So the semiperimeters are the odd squares ≥ 9 and the numbers $2u^2$ with $u \ge 2$, that is, by the characterization of A096033, the elements of $\mathcal{L}$ other than 1 and 2. This is Theorem 3, and Theorem 2 follows by counting, as above.

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A308222.lean` (172 lines) of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A308222.lean` without modifying it, proves `theorem result : statement` in the namespace `Math0.Frontier.A308222`, and ends with

```
#math0_settles Math0.Frontier.A308222.result Math0.Statements.A308222.statement
```

Lemmas 5 and 6 are in the statement file. Lemmas 7, 8, 9 and 11, Corollary 10 and `infinite_L` are private declarations of the proof file, named in Section 3. The proof file contains two small evaluations by `decide` (the base case of Lemma 11 and $\mathrm{count}_{\mathcal{L}}(2) = 1$). The proof term that `decide` produces is checked by the kernel, which reduces the decidability instances; these instances are defined through `decidable_of_iff` from Boolean searches proved equivalent to the predicates (`perimB_iff`, `legHypB_iff`). The kernel data checks of the statement file use `decide +kernel`. Neither file uses `native_decide`, `sorry`, `set_option`, or a declared axiom.

Audit. `#math0_settles` (in `lean/Math0/Audit.lean`) fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound`, which excludes `sorryAx`, the axioms behind `native_decide`, and custom axioms; and (3) the type of `result` is syntactically `statement` or its negation, up to metadata, with no unfolding. Running `lake env lean Math0/Frontier/OEIS/A308222.lean` in `lean/` (about 10–20 seconds) reports

```
math0 audit: 'Math0.Statements.A308222.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A308222.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A308222.result' proves 'Math0.Statements.A308222.statement'
```

Toolchain and reproduction. We use Lean 4 [6] `v4.33.1` and Mathlib [5] `v4.33.1`, pinned in `lean/`. The repository also pins formal-conjectures [2] at commit `c77fdd4`; this conjecture is not in that collection, so math0 formalized it. To reproduce, run `scripts/verify.sh` from the repository root: it builds all math0 results with `lake build`, which re-runs every audit command, and checks that the pinned dependencies are unmodified. Building the statement file alone (`lake build Math0.Statements.OEIS.A308222`) takes about two minutes, almost all of it in the kernel data checks. Independently of the formal proof, the referees checked numerically that the perimeters up to 8000 are 74 numbers, of which the first 46 are the listed terms, and that they satisfy $a(n) = 2 \cdot \mathrm{A096033}(n+2)$ for all 74 indices.

References

[1] J. R. Carlson, *Determination of Heronian triangles*, Fibonacci Quart. **8** (1970), no. 5, 499–506, 551.

[2] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An open and evolving benchmark for verified discovery in mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[3] George (Mathematics Stack Exchange user), answer to *Semiperimeter of isosceles Heronian triangles*, Mathematics Stack Exchange, May 17, 2019, `https://math.stackexchange.com/a/3230072`.

[4] P. Kagey, *Semiperimeter of isosceles Heronian triangles*, Mathematics Stack Exchange, question 3230019, May 17, 2019, `https://math.stackexchange.com/q/3230019`.

[5] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[6] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[7] The OEIS Foundation Inc., *Difference between leg and hypotenuse in primitive Pythagorean triangles*, `https://oeis.org/A096033`, entry by L. Beedassy (Jun. 16, 2004).

[8] The OEIS Foundation Inc., *Numbers that are the perimeter of a primitive Heronian isosceles triangle*, `https://oeis.org/A308222`, entry by P. Kagey (May 16, 2019), conjecture added May 17, 2019; revision #20, Aug. 12, 2022.

[9] D. Singmaster, *Some corrections to Carlson's "Determination of Heronian triangles"*, Fibonacci Quart. **11** (1973), no. 2, 157–158.

[10] Wikipedia, *Integer triangle*, section "Isosceles Heronian triangles", `https://en.wikipedia.org/wiki/Integer_triangle`, accessed October 2026.